

Munjal Nayak

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CAREER PROFILE

- Pursuing Master of Engineering Science in Electrical Engineering @ **UNSW**
- Former ETCS Application Engineer with 2 years of experience @ **ALSTOM**
- Completed B.Tech in Electrical Engineering with 8.71 CGPA @ **PDEU**
- Strong hands-on expertise in PCB design, hardware bring-up, and embedded firmware development using ESP32, STM32, Embedded C, and Python.
- Effective communicator with a strong ownership mindset, capable of working independently or in high-responsibility engineering environments

Education

Master of Engineering Science (Electrical)

Specialization: Embedded Systems

University of New South Wales (UNSW)

2026 – Present

Expected Grad: 2028

B.Tech in Electrical Engineering

CGPA: 8.71 / 10.0

Pandit Deendayal Energy University (PDEU)

2019 – 2023

2 Publications 10/10 in Final Project

- Authored **2 research papers** on machine learning applications in electrical engineering.

Relevant Experience

ETCS Application Engineer (Onboard Systems)

Bengaluru, India

ALSTOM

2023 – 2025

SIL4 Railway Signalling Onboard Systems Data Configuration

- Configured and validated onboard ETCS data for **3 European Vital Computer platforms** (EVC3, EVC2 Plus, EVC2), ensuring 100% compliance with SIL4 safety standards.
- Reduced data configuration errors by implementing systematic validation procedures across **2 major railway projects** serving 50,000+ daily passengers.
- Authored technical documentation and conducted knowledge transfer sessions, enabling team scalability and reducing onboarding time for new engineers.
- **Key Projects:**
 - *NCRTC MRTS (Delhi)* — EVC3 onboard data engineering and end-to-end validation
 - *Madrid ETCS Programme* — EVC2/EVC2P configuration data preparation and integration

Licenses

Unrestricted Car and Rider license with access to own car, First Aid & CPR, Heavy Rigid (learner), Forklift and Order Picker license, White Card, Blue Card (WWCC).

Languages

IELTS overall band 7 (English).

Additional Experience

Data Cable Technician

Sydney, Australia

GM Cables

Dec 2025 – Present

Cat6A Fiber Optics Structured Cabling Network Infrastructure

- Installing and certifying CAT5e/CAT6/CAT6A and fiber optic cabling for enterprise clients, maintaining **100% TIA/EIA compliance**.
- Deploying structured cabling infrastructure for schools, commercial offices, and telecommunications facilities across Sydney metropolitan area.
- Managing cable pathways and maintaining optimal bend radii while adhering to OSHA safety standards in active construction environments.

IoT Developer Intern (Hardware & Firmware)

Gandhinagar, India

Eleics Design

2023

ESP32 Firmware IoT Product Development

- Developed production-ready embedded firmware for ESP32-based IoT products using ESP-IDF, contributing to **2 commercial product releases**.
- Led firmware development for “WiDrive” product, including PCB design input, protocol implementation, and system integration testing.
- Implemented industrial communication protocols (SPI, UART, MQTT, HTTP, USB) with **<100ms latency** for real-time IIoT applications.
- Collaborated in design reviews and test planning, improving product reliability through systematic debugging and incremental enhancements.

Skills

Core Hardware

PCB and schematic design, component derating and selection, PCB manufacturing processes and DFM/DFT considerations, soldering and SMD rework, oscilloscope and mixed-signal debug, signal integrity awareness, power electronics (multilevel inverters, motor drives, DC–DC conversion), rapid prototyping (3D printing, breadboarding), hardware bring-up and reverse engineering.

Embedded

Embedded C for bare-metal and MCU environments, ESP32 (ESP-IDF), STM32, Python for scripting and test harnesses, UART/SPI/I²C peripheral integration, GPIO and timer configuration, MQTT/MQTTS and HTTP/Web APIs, basic RTOS concepts (tasks, scheduling, synchronization), debugging with logic analyzers and serial consoles, firmware over-the-air (basic exposure).

Systems & Networking

OPNsense, pfSense, WireGuard, Wireshark, AdGuard, EJBCA, Proxmox VE, Docker, TrueNAS, PostgreSQL, MariaDB, Ubuntu Server 24.04, Rocky Linux, Windows Server 2022/2024, Active Directory (AD), AD CS, Azure Active Directory, Azure IAM, basic PKI and certificate management, network segmentation and secure remote access.

Safety, Quality & Standards

Working knowledge of SIL4 safety concepts in railway signaling, configuration management, traceability, version control with Git/GitHub, test planning and documentation, basic understanding of EMC/ESD concerns and environmental testing constraints.

Product & Project

Requirements elicitation and clarification, translating customer and operator needs into technical specifications, writing and reviewing technical documentation, coordinating with cross-functional and multi-disciplinary teams, basic project planning and effort estimation, stakeholder communication and expectation management.

Soft Skills

Clear written and verbal communication, structured problem solving, ownership mindset, ability to work independently or within cross-functional teams.

Certifications

- IBM – Introduction to Cloud Computing
- Microsoft – Create Machine Learning Models in Microsoft Azure
- Red Hat – Fundamentals of Red Hat Enterprise Linux
- MathWorks – Power Electronics Simulation Onramp
- IBM – Getting Started with Git and GitHub
- Keysight Technologies – Essential RF Power Measurements
- IIT Bombay – Understanding Incubation and Entrepreneurship
- UC Irvine – The Raspberry Pi Platform and Python Programming
- Coursera – Docker for Beginners

Personal Projects

IoT Smart Panel (Industrial Motor Control)

ESP32 VFD Industrial IoT

- Designed an ESP32-based IoT controller to manage a three-phase squirrel cage induction motor via VFD.
- Implemented cloud connectivity for data logging and predictive analysis, enabling remote monitoring and control of motor performance.

IoT Smart Irrigation System

ESP32 Home Assistant Embedded Sensors

- Developed an ESP32 + Home Assistant-based smart irrigation solution for soil moisture and temperature monitoring.
- Enabled per-plant automated watering based on sensor feedback, improving water efficiency and reducing manual intervention.

Low Power Multilevel Inverter

Power Electronics STM32

- Implemented a 5-level H-bridge inverter (+24, +12, 0, -12, -24) using 5 MOSFETs and an L-type power supply, controlled by an STM32 microcontroller.
- Focused on switching strategy, efficiency, and safe operation under low-power conditions.

Publications

- ELSEVIER – *Engineering Applications of Artificial Intelligence*: “An explainable artificial intelligence based approach for the prediction of key performance indicators for a 1 megawatt solar plant under local steppe climate conditions.”
- International Conference on Sustainable Energy and Clean Technologies 2022 – “Using machine learning techniques to predict the effect of different parameters on inverter efficiency.”

Final-year Project

Electrical Efficiency Optimization in the Industrial Sector: An Embedded Systems Approach to Real-time Asset Monitoring and Predictive Control

- Designed and implemented an embedded system for real-time monitoring of industrial assets, enabling data-driven efficiency improvements and predictive maintenance.

References

References provided on request.